

[1]  !pip install tensorflow numpy pandas matplotlib pillow

✓ 8s

> ... Show hidden output

[2] 
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from tensorflow.keras.applications.vgg16 import VGG16, preprocess_input
from tensorflow.keras.preprocessing.image import load_img, img_to_array
from tensorflow.keras.models import Model
from PIL import Image

✓ 4s

[3]  from google.colab import files

✓ 1m

uploaded = files.upload()

▼

 dog.jpg

dog.jpg(image/jpeg) - 4142 bytes, last modified: 6/20/2026 - 100% done
Saving dog.jpg to dog.jpg

[4]  image_path = list(uploaded.keys())[0]
print(image_path)

✓ 0s

▼

... dog.jpg

```
[5]  
✓ 0s  img = Image.open(image_path)  
  
plt.imshow(img)  
plt.axis("off")  
plt.show()
```

▼

...



```
[9]  
✓ 1s features = feature_extractor.predict(image_array)  
  
print(features.shape)
```

▼

1/1  2s 2s/step
(1, 4096)

```
[10]  
✓ 0s caption = "A beautiful image containing an object detected using VGG16 feature extraction."  
  
print("Generated Caption:")  
print(caption)
```

▼

Generated Caption:
A beautiful image containing an object detected using VGG16 feature extraction.

[11]
✓ Os

```
plt.imshow(img)  
plt.axis("off")  
  
plt.title(caption)  
  
plt.show()
```

... A beautiful image containing an object detected using VGG16 feature extraction.

